Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech
Code: 110406

Subject: Signals and Systems

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) A system is BIBO stable if
- (i) Input is bounded
- (ii) Output is bounded
- (iii) Bounded input gives bounded output
- (iv) Output is zero

(b) Given $h(t) = \delta(t-1)$, and $x(t) = \sin(t)$, then $y(t)$ ....
- (i) $\sin(t)$
- (ii) $\sin(t+1)$
- (iii) $\sin(t-1)$
- (iv) 0

(c) The complex exponential $x(t) = e^{j\omega t}$ is
- (i) Non-periodic
- (ii) Periodic
- (iii) Energy signal
- (iv) Random

(d) The convolution integral is given by
- (i) $\int x(t)\,h(t)\,dt$
- (ii) $\int x(\tau)\,h(t+\tau)\,d\tau$
- (iii) $\int x(\tau)\,h(t-\tau)\,d\tau$
- (iv) $\int x(t-\tau)\,h(-\tau)\,d\tau$

(e) Which of the following is an example of a deterministic signal?
- (i) White noise
- (ii) Gaussian noise
- (iii) Random signal
- (iv) Sinusoidal signal

(f) The signum function can be written as
- (i) $u(t) - 1$
- (ii) $1 - u(t)$
- (iii) $2u(t) - 1$
- (iv) $u(t) + u(-t)$

(g) Fourier series for even symmetry in the waveform leads to
- (i) Only sine terms
- (ii) Only cosine terms
- (iii) Both sine and cosine
- (iv) DC term only

(h) Stability requires the ROC to include
- (i) Origin
- (ii) Unit circle
- (iii) Infinity
- (iv) Imaginary axis

(i) Parseval's theorem relates
- (i) Time shift and frequency shift
- (ii) Magnitude and phase
- (iii) Energy in time and frequency domains
- (iv) Causality and stability

(j) The integral $\int_{-\infty}^{\infty} X(\omega)\delta(\omega - \omega_0)d\omega = \cdots$
- (i) $X(0)$
- (ii) $X(\omega)$
- (iii) $X(\omega_0)$
- (iv) 0

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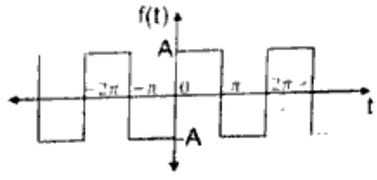
Q.2 (a) Find the Fourier transform of the signal $x(t) = e^{-|t|}u(t)$ and plot its magnitude and phase spectrum [7]

(b) Solve the differential equation [7]

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 5y(t) = 5x(t)$$

With initial condition $y(0) = 1$ and $\frac{dy(t)}{dt}\Big|_0 = 2$ and input $x(t) = u(t)$

Q.3 (a) Obtain the exponential Fourier series for the wave form shown below, and draw the frequency spectrum [7]



(b) Determine the value of signal x(n) at n = 0 and n = ∞ if [7]

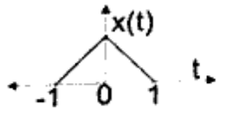
$$X(z) = \frac{2z^2 + 0.25}{(z + 0.25)(z - 1)}$$

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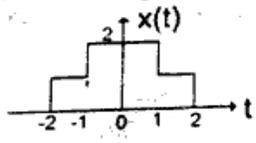
Q.4 (a) Find the convolution of two continuous-time signals [7]

$$x(t) = e^{-4t}u(t) \text{ and } h(t) = u(t - 3)$$

(b) Sketch the following signals x(-2t + 1) [7]



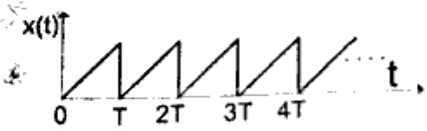
Q.5 (a) Determine the Fourier transform of the signal shown below [7]



(b) Find the z-transform and ROC of the sequence [7]

$$x(n) = \left(\frac{-1}{3}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(-n - 1)$$

Q.6 (a) Find the Laplace transform of the signal x(t) as shown below [7]



(b) Check whether the signal is periodic or not. If periodic, then find its period. [7]

$$x(n) = 2e^{j\left(3\pi n + \frac{\pi}{4}\right)}$$

Q.7 (a) Check whether the system is linear/non-linear and time-variant/time-invariant [7]

$$y(t) = t.x(t)$$

(b) Find the state space model of the system represented by the differential equation: [7]

$$\frac{d^3y(t)}{dt^3} + 7\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = 9x(t)$$

Q.8 (a) Find the DFT of the sequence $x(n) = \{1,1,1,1\}$. [7]

(b) Find the Laplace Transform of $x(t) = te^{-2t}u(t)$. [7]

Q.9 Write short note on *any two* of the following: [7 × 2 = 14]

(a) Properties of z-transform

(b) Parseval's theorem with its application

(c) Sampling Theorem

(d) Power and energy signals

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